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Courtesy of Professor William Millhiser

OPM 3730

Assignment 5: Process Simulation Modeling and Analysis[1]

Objective. This assignment gives you the opportunity to model a service operation using business process simulation software. It is taken from the American Red Cross's improvement initiative which was published in a journal on decision analytics, the *Journal on Applied Analytics*, formerly called *Interfaces.*[2] You will learn (i) how to build a simulation model from scratch, (ii) how to run and interpret the outputs of a simulation model, and (iii) how to edit the simulation model to do "what-if" analysis to consider the likely impact of proposed process improvements.

1 Process description: American Red Cross Mobile Blood Clinics

Peggy Ladowski, a regional director of donor services for the American Red Cross, was wondering how to improve services to blood donors. She knew that donor satisfaction was one important factor in the American Red Cross's ability to attract a continuing stream of blood donors. Recently she had heard an increasing number of complaints that donors were being held up in long lines at the organization's mobile blood collection operations, commonly termed bloodmobiles. Ladowski was considering several design alternatives to reduce these complaints.

1.1 American Red Cross

The American Red Cross operated over 400 fixed and mobile collection sites in 38 regions, such as the one for which Ladowski worked. Fixed sites were permanently established locations open during regular hours. In contrast, mobile sites were usually operations set up using portable equipment at a temporary site provided by a

The American Red Cross depended heavily

[1] This assignment was prepared by William P. Millhiser for the Zicklin School and is copyright by William P. Millhiser, and may not be distributed, shared, or posted in any way without explicit written consent.

[2] The article is: J.E. Brennan, B.L. Golden, & H.K. Rappoport (1992), "Go with the Flow: Improving Red Cross Bloodmobiles Using Simulation Analysis," *Interfaces*, 22(5), pp. 1-13, available on Blackboard → OPM 3730 → Assignments. **Figure 1** and **Table 1** below were adapted from Brennan et al. (1992) by W.P. Millhiser. The Brennan et al. paper was the basis for R.D. Klassen and J. Haywood-Farmer's (2005), "Red Cross Mobile Blood Clinics—Improving Donor Service," in *Cases in Operations Management: Building Customer Value Through World-Class Operations* (edited by R.D. Klassen & L.J. Menor, Sage Publications), pp. 101-105. The text in §1 was excerpted from Klassen and Haywood-Farmer (2005).

sponsor, often for a few days during a blood drive. Mobile sites accounted for about 80 percent of the collected blood.
on regular repeat donors; recruiting and retaining them was a challenge. Surveys of donors indicated that time spent at the clinic and time spent waiting in line were important determinants of their level of satisfaction and significantly influenced their decision to return. Donor service time was also important to sponsors, especially employers who gave employees time off to donate blood. Although the American Red Cross told donors and sponsors that donation required about one hour, for many, it took one and a half to two hours.

Several factors contributed to the delay. Because the American Red Cross did not use an appointment system for its blood clinics, arrival at blood donation clinics was essentially random. In addition, the recent increase in awareness that blood could transmit some serious diseases had affected the collection process in two ways. Donors had to read more material, answer more questions and complete a confidential self-exclusion step. Second, the threat of disease transmission had forced staff to use gloves, needle guards and other devices and procedures to reduce their exposure to blood.

Despite an increase in the complaints from sponsors and donors about the time spent at blood clinics, unfortunately, the American Red Cross had been able to dedicate few additional resources to alleviate the problem as it budgets were limited and nurses were in short supply.

1.2 Blood collection process

Although many American Red Cross regions had developed their own specific variations, all had to perform the same basic procedures as described in **Figure 1**. First, a registrar asked all arriving prospective donors for certain demographic information, wrote this on a donor registration form, gave prospective donors reading material and directed them to the waiting area. Prospective donors then read the information provided and answered the questions on a self-administered health history (SAHH) form with yes-no answers.

Third, a collections staff member recorded the prospective donor's temperature, blood pressure and pulse rate, took a blood sample and tested it for haemoglobin level. A second iron level test was given those who failed the haemoglobin test. Prospective donors whose vital signs fell outside standard limits were rejected.

Fourth, a collections staff member clarified the prospective donor's responses to the SAHH, if necessary, and asked him or her a series of health history questions. The purpose of this step was to ascertain the likelihood that the prospective donor carried HIV hepatitis virus. If any of this information indicated that the prospective donor should not donate blood, he or she was rejected. To accepted donors, the staff member issued and explained the confidential unit exclusion (CUE) form that allowed any donor to indicate confidentially whether his or her blood should be used. This procedure allowed a donation resulting from peer pressure to be quarantined and later discarded.

Next, a staff member issued a blood bag set to the donor. The bag set, the registration form and the CUE form were all bar coded. The donor then completed the CUE form in a private area.

For the donation itself, the donor lay on a bed while a phlebotomist disinfected his or her arm, inserted a needle and drew the blood. After about 450 ml of blood had been drawn, the phlebotomist withdrew the needle and applied a bandage to the donor's arm. Each phlebotomist was assigned to serve several donor beds.

Finally, following donation, the donor sat and recovered in the canteen, Which provided some liquid refreshment and a snack. This step allowed staff to monitor donors to make sure they had no immediate post-donation reactions.

1.3 Data collection

To better understand the problem, Ladowski commissioned the collection of certain data using a six-bed clinic model, which was the most common one used. Such a clinic typically served 50 to 75 donors over a six-hour period. She believed that understanding it thoroughly would also help her deal with other models. Historical records and standard operating procedures gave data on donor flow, percentages and locations of donor deferrals. Data were also collected at several clinics on the frequency of donor arrivals and the time required for each major step.

1.4 Results

The observed service times and standard deviations for the various stages of the donor process are summarized in **Table 1**. This allows a high level analysis of process capacity (Table 2) which suggests that the 2 phlebotomists are the process bottleneck, limiting hourly flow to at most 15 donors. The data also revealed several interesting patterns for the arrivals of potential donors. The vast majority of the clinics, approximately 91 percent, experienced one of three basic arrival patterns. Approximately 22 percent of the clinics — dominated by open, community drives in which donors must come in their own time — showed one steep peak near midday (**Figure 2**). Thirty-six percent of the clinics displayed both mid-morning and mid-afternoon peaks. This distribution was most prevalent in employer-sponsored clinics in which employees got time off work to donate. The remaining onethird of the clinics showed a single large peak during the morning hours of the clinic, followed by a gradual decline in the arrival rate. This pattern was most common in colleges or facilities with shift work.

1.5 Conclusion

With these data in hand, Ladowski set about trying to identify possible options for improved service. She decided to begin her work by examining arrivals that displayed the midday peak (**Figure 2**).

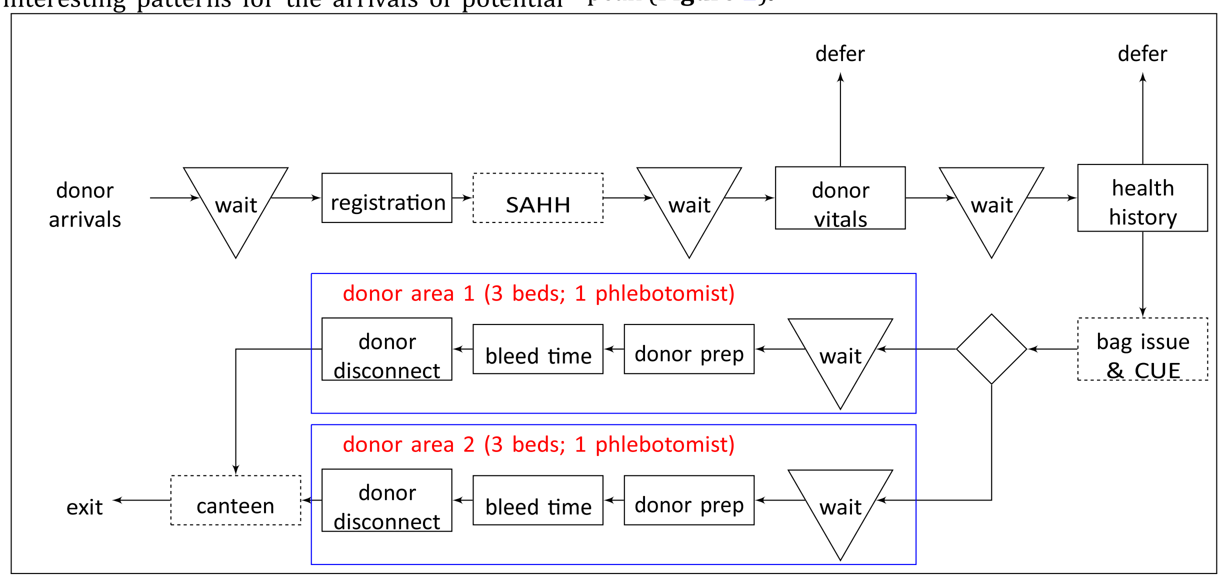


Figure 1: The 6-bed Red Cross mobile blood clinic donation process. Solid boxes denote capacity-limited steps; dashed boxes represent capacity-unlimited steps.

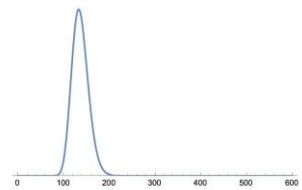
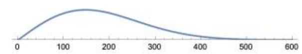
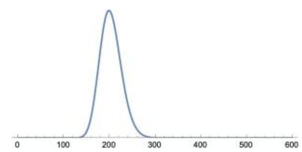
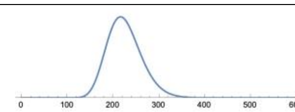
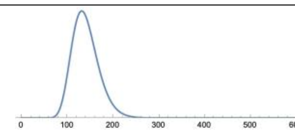
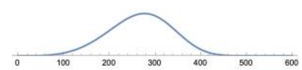
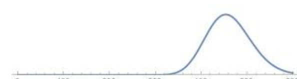
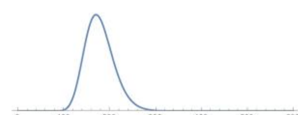
Step	No. of staff	Mean (seconds)	Std. dev. (seconds)	Distribution	AnyLogic delay time parameters (seconds)	PDF
Registration	1	144.4	54.3	Lognormal	lognormal(4.906,0.132,0)	
SAHH	—	184.9	96.9	Weibull	weibull(2.0,209.5,0)	
Donor vitals	2	213.7	74.8	Lognormal	lognormal(5.307,0.116,0)	
Health history	2	241.6	102.8	Lognormal	lognormal(5.404,0.166,0)	
Bag issue & CUE	1*	151.9	72.4	Lognormal	lognormal(4.921,0.205,0)	
Donor prep	2**	267.8	65.6	Weibull	weibull(4.4,293.2,0)	
Bleed time	2**	484.1	162.6	Lognormal	lognormal(6.129,0.107,0)	
Disconnect	2**	191.6	84.3	Lognormal	lognormal(5.167,0.177,0)	
Canteen	1*	861.0	509.4	Lognormal	lognormal(6.608,0.300,0)	 †

Table 1: Red Cross mobile blood clinic service time distributions. (Source: Brennan et al. 1992.)

* Staff are involved for only a small fraction of this time.

** Phlebotomists perform the preparation and disconnect steps; they need to monitor the donor only periodically during the bleed step.

† The horizontal scale for the Canteen time is different from others for space considerations.

Resource Pool	No. of resource units	Avg. unit load (seconds per donor)*	Resource pool capacity (donors / hr)	Total daily capacity (donors per 6 hours)	
Registrar	1	144.4	24.9	149.6	
Donor Vitals Staff	2	213.7	33.7	202.2	
Health History Staff	2	241.6	29.8	178.8	
Phlebotomist	2	459.4	15.7	094.0	← process bottleneck
Donor Beds	6	943.5†	22.9	137.4	

Table 2: High-level average capacity analysis, ignoring queueing delays.

* Source: Table 1.

† Ignores delay incurred by occupied beds waiting for phlebotomist.

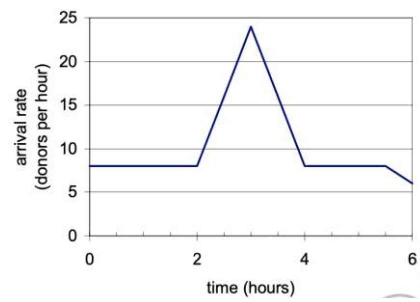


Figure 2: The average number of arrivals per hour for a typical "midday peak" demand (source: Klassen and Haywood-Framer, 2005, Fig. 3a).

2 Assignment Instructions

For this assignment you will build (and submit) 4 AnyLogic .alp simulation models. Each is described below.

2.1 Model 1: Simulate the basic process

Build a simulation model of a typical a day with a "midday peak" described above. Your process simulation model should have enough KPIs to answer the following questions (the

KPIs from the Cleveland Clinic examples from class meetings 20 and 21 will suffice). Save a copy of this model as Red Cross (base model).alp and answer the following.

1. What is the total average flow time for all donors? What is the range of flow times for individual donors (min and max)?

2. On average, how long does a donor wait in queues? What fraction is this of the total time in the system?

3. Where in the bloodmobile is the queueing the worst? Is this a bottleneck? Explain.

2.2 Model 2: Pooling staff (combining two steps)

One alternative is to change from a series of individual steps (with individual staff assigned to each step) to a pool of staff who process donors through both steps. Thus, the two staff formerly assigned to Donor Vitals and the two staff formerly assigned the Health History form a staff pool of four. Each staff member then completes both steps for each donor, eliminating the in-between delay. Assume that the Donor Vitals and the Health History steps have the same probability distributions described in **Table 1** (i.e., we will continue to model them as 2 steps, each with their own service time probability distributions). Modify your model to reflect this improvement, save it as Red Cross (combined steps).alp, and run it to answer the following questions.

1. On average, how much has the total wait time changed for each donor (or alternatively, how much has the flow time improved for the average donor)?

2. Why did combining these two steps using a pool alter the average donor's wait time?

2.3 Model 3: Combining donor bed areas

Revert to the base model you developed in §2.1. Rather than having each phlebotomist focus on a designated bed area (i.e., 3 beds each), another option is to have a single donor bed area with 6 beds served by both phlebotomists. Modify your model to reflect this improvement, save it as Red Cross (combined areas).alp, and run it to answer the following questions.

1. On average, how much wait time has been eliminated for each donor (or alternatively, how much has the flow time improved for the average donor)?

2. What fraction of time does the average donor spend waiting in queue?

3. Why was this change in §2.3 so much larger than combining two steps in §2.2?

2.4 Self-registration and a "floater"

Revert to the base model you developed in §2.1. A last alternative is to change the process to require the donors, most of whom were repeat customers, to self-register themselves and then complete the Self-Administered Health History (SAHH) form on their own immediately after arrival. Doing so frees that staff member (formerly, the registrar) to be a "floater" who can assist where the help is needed most. Assume that the floater can perform any step in the process, including phlebotomy. Modify your model to reflect this improvement, save it as Red Cross (self-reg and floater).alp, and run it to answer the following questions.

1. On average, how much total wait time has been eliminated for the average donor (or alternatively, how much has the flow time improved for the average donor)? Is there a likely process bottleneck now? Explain.

2. What fraction of time does the donor spend waiting in a queue?

3. Why was this change (§2.4) so much larger than any of the previous changes (§§2.2-2.3)?

4. Could you expect to see further improvement if all three alternatives were combined (i.e., pooling staff, combining donor bed areas, and self-registration with a floater)?